

# Ayoub Foussoul

ayoub.foussoul@chicagobooth.edu · www.ayoubfoussoul.com

## ACADEMIC APPOINTMENTS

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**University of Chicago Booth School of Business**

*Postdoctoral Researcher*

2025 –

*Chicago, IL*

## EDUCATION

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**Columbia University**

*PhD in Operations Research*

2020 – 2025

*New York, NY*

- Advisor: Vineet Goyal

- Thesis: Robust and Tractable Policies for Resource Allocation Under Uncertainty

*Master of Science in Operations Research*

**École Polytechnique**

*Master of Science in Applied Mathematics*

2017 – 2020

*Paris, France*

*Bachelor of Science in Mathematics and Computer Science*

**Lycée Moulay Youssef**

*Classes préparatoires, Mathematics, Physics, Computer Science*

2015 – 2017

*Rabat, Morocco*

## PAPERS

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- [1] Discrete Input Landscapes of Neural Networks ([link](#))  
*Ozan Candogan & Ayoub Foussoul*  
*Working paper*
- [2] Neural Scaling Laws for Customer Choice Prediction  
*Ozan Candogan, Ayoub Foussoul & Feiyu Han*  
*Working paper*  
*Preliminary version accepted at the **NeurIPS 2026 MLxOR Workshop***
- [3] Deep Flow Networks: Optimization-Friendly Surrogates for Integer Predict-then-Optimize ([link](#))  
*Ozan Candogan & Ayoub Foussoul*  
*Submitted to **Management Science***  
*Preliminary version appeared in proceedings of **ICML 2026 (spotlight paper)***
- [4] Distributionally Robust Newsvendor on a Metric ([link](#))  
*Ayoub Foussoul & Vineet Goyal*  
*Major revision in **Operations Research***  
*Extended abstract appeared in proceedings of **EC 2025***
- [5] LP-based Approximations for Disjoint Bilinear and Two-Stage Adjustable Robust Optimization ([link](#))  
*Omar El Housni, Ayoub Foussoul & Vineet Goyal*  
***Mathematical Programming***  
*Preliminary version appeared in proceedings of **IPCO 2022***
- [6] Two-Stage Stochastic Stable Matching ([link](#))  
*Ayoub Foussoul, Yuri Faenza & Chengyue He*  
*In proceedings of **IPCO 2024***

- [7] Minimum Cut Representability of Stable Matching Problems ([link](#))  
*Ayoub Foussoul, Yuri Faenza & Chengyue He*  
**Major revision in *Operations Research***  
*★ Honorable Mention, INFORMS Optimization Society, 2025 Student Paper Prize*
- [8] Fully-Dynamic Load Balancing ([link](#))  
*Ayoub Foussoul, Vineet Goyal & Amit Kumar*  
***Mathematical Programming***  
*Preliminary version appeared in proceedings of **IPCO 2024***
- [9] Last Switch Dependent Bandits with Monotone Payoff Functions ([link](#))  
*Ayoub Foussoul, Vineet Goyal, Orestis Papadigenopoulos & Assaf Zeevi*  
*In proceedings of **ICML 2023***
- [10] MNL-Bandit in Non-Stationary Environments ([link](#))  
*Ayoub Foussoul, Vineet Goyal & Varun Gupta*

## TECHNICAL REPORTS

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- [1] Conditioning of Piecewise Linear Approximations with an Application to the Monitoring of the COVID-19 Epidemic  
*Marianne Akian, Jérôme Bolt, Ayoub Foussoul, Stephane Gaubert & Laurent Massoulié*  
*★ École Polytechnique Research Prize in Applied Mathematics*
- [2] Identification of the Optimal Tropical Hypersurfaces Explaining Consumers' Responses to Prices  
*Xavier Allamigeon, Ayoub Foussoul & Stephane Gaubert*  
*★ École Polytechnique Research Prize in Applied Mathematics*

## TEACHING EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Optimization I, IEOR E6613</b><br><i>Core PhD course on linear and convex optimization</i><br><i>Teaching Assistant &amp; Guest Lecturer</i>                   | Fall 2021, 2022         |
| <b>Optimization II, IEOR E6614</b><br><i>Core PhD course on combinatorial optimization</i><br><i>Teaching Assistant</i>                                           | Spring 2023, 2024, 2025 |
| <b>Convex Optimization, EEOR E6616</b><br><i>PhD course on convex optimization</i><br><i>Teaching Assistant</i>                                                   | Spring 2022             |
| <b>Optimization Methods, IEOR E4004</b><br><i>Core master's course on optimization</i><br><i>Teaching Assistant &amp; Guest Lecturer</i>                          | Fall 2023               |
| <b>Applications for Financial Engineering, IEOR E4500</b><br><i>Master's course on quantitative methods in financial engineering</i><br><i>Teaching Assistant</i> | Spring 2021             |
| <b>Academic Tutor, Le Valdocco</b><br><i>Provided academic support and mentoring to underprivileged students at a non-profit educational organization.</i>        | 2017 – 2018             |

## INDUSTRY EXPERIENCE

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### INRIA

Mar 2020 – Sep 2020

*Research Intern*

*Paris, France*

- **Constrained piecewise-linear approximation for epidemic monitoring:** Developed a dynamic-programming method for constrained piecewise-linear approximation in  $L_p$  norms ( $p \geq 1$ ) and applied it to French COVID-19 observables (SIVIC/SAMU) (see acknowledgments in paper).
- **VC dimension of tropical polynomials:** Studied tropical polynomial classifiers and proved that the VC dimension of tropical polynomials matches that of classical real polynomials.

### Padoa

Jun 2019 – Sep 2019

*Data Analyst Intern*

*Paris, France*

Occupational-medicine platform connecting clinicians, employers, and employees

- Implemented a Health Statistics page in the clinician-facing web app.
- Extended the backend Stats API to expose the KPI/statistics required by the new dashboard.

## SERVICE AND OUTREACH

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Elected Student Representative, École Polytechnique (Class of 2020)

Co-organizer of IEOR PhD Tutorial Series

Reviewer: Management Science, Mathematical Programming, Optimization Letters,  
NeurIPS MLxOR Workshop, INFORMS Journal on Computing, IPCO, SODA, STOC, EC, ICML

## AWARDS

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|---------------------------------------------------------------------------|------|
| Honorable Mention in the INFORMS Optimization Society Student Paper Prize | 2025 |
| Postgraduate Excellence Scholarship (OCP Foundation)                      | 2020 |
| École Polytechnique Research Prize in Applied Mathematics                 | 2020 |
| French Government Major-Excellence Scholarship                            | 2017 |

## TALKS

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*Deep Flow Networks: Optimization-Friendly Surrogates for Integer Predict-then-Optimize*

- INFORMS Annual Meeting, November 2026, San Francisco, CA
- LAMP Workshop at TTIC, August 2026, Chicago, IL (poster session)
- Machine Learning for Algorithms Workshop at STOC 2026, June 2026, Salt Lake City, UT (poster session)
- Chicago Operations Day, June 2026, Chicago, IL (poster session)

*Distributionally Robust Newsvendor on a Metric*

- INFORMS Annual Meeting, October 2025, Atlanta, GA
- Revenue Management and Pricing Conference (RMP), July 2025, New York, NY
- ACM Conference on Economics and Computation (EC), July 2025, Stanford, CA
- Manufacturing & Service Operations Management Conference (MSOM), June 2025, London
- Columbia University Data Science Day, April 2025, New York, NY (poster session)
- IEOR Colloquium, November 2024, New York, NY
- INFORMS Annual Meeting, October 2024, Seattle, WA
- Cornell ORIE Young Researchers Workshop, October 2024, Ithaca, NY

- Northwestern Kellogg Operations Management Rookiepaloosa, October 2024, Evanston, IL

*Two-Stage Stochastic Stable Matching*

- Integer Programming and Combinatorial Optimization Conference (IPCO), July 2024, Wrocław
- Columbia University Data Science Day, April 2024, New York, NY (poster session)

*Fully-Dynamic Load Balancing*

- Integer Programming and Combinatorial Optimization Conference (IPCO), July 2024, Wrocław
- Columbia IEOR Student Seminar, February 2024, New York, NY

*Last Switch Dependent Bandits with Monotone Payoff Functions*

- International Conference on Machine Learning (ICML), July 2023, Honolulu, HI (poster session)

*MNL-Bandit in Non-Stationary Environments*

- DSI Financial and Business Analytics Poster Session, November 2023, New York, NY (poster session)
- INFORMS Annual Meeting, October 2023, Phoenix, AZ
- Revenue Management and Pricing Conference (RMP), July 2023, London

*LP-based Approximations for Disjoint Bilinear and Two-Stage Adjustable Robust Optimization*

- International Symposium on Mathematical Programming (ISMP), July 2024, Montréal
- International Research and Innovation Seminar (IRIS), December 2023, Ben Guerir
- INFORMS Annual Meeting, October 2022, Indianapolis, IN
- International Conference on Continuous Optimization (ICCOPT), July 2022, Bethlehem, PA
- Integer Programming and Combinatorial Optimization Conference (IPCO), June 2022, Eindhoven
- INFORMS Optimization Society Conference (IOS), March 2022, Greenville, SC